Ankita Kotangale

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Research Profile

Ph.D. student in Learning Design & Technology at Purdue University whose research focuses on understanding how Design Thinking develops through educational experiences, how that development can be meaningfully assessed, and how learning experiences can be designed to support it. Research interests include Design Thinking development and assessment, informal STEM learning, instructional design, and human-AI collaboration in Design Thinking.

Education

Ph.D. in Learning Design & Technology — Purdue University, West Lafayette, IN | Expected Summer 2027

Master of Science in Technology Management — Pittsburg State University, Pittsburg, KS | 2020

Bachelor of Engineering in Information Technology — S.B.J.I.T.M.R., Nagpur, Maharashtra, India | 2018

Research Interests

Design Thinking development; Design Thinking assessment and instrument validation; informal STEM learning; engineering and computing education; design of learning experiences; human-AI collaboration and learner agency.

Research Experience

Lead Researcher — GERI Design Thinking Study — Purdue University | 2023–Present

- Examined changes in high school students' Design Thinking mindset and demonstrated application during a two-week informal STEM learning experience.
- Adapted a self-report Design Thinking mindset instrument and developed a scenario-based assessment and scoring rubric.
- Conducted pre/post quantitative analyses, reliability analyses, and interpretation across self-report and scenario-based measures.
- Prepared a first-author manuscript and conference presentation based on study findings.

Researcher — FIRST Robotics Design Thinking Study — Purdue University | 2026–Present

- Examined Design Thinking mindset and scenario-based application among students participating in robotics learning environments.
- Administered Design Thinking assessments, scored scenario responses, and conducted quantitative analyses.
- Used the study to extend Design Thinking assessment across an additional informal STEM context.

Researcher — NSF-Funded DEAP Project — Purdue University | 2023–2024

- Investigated computing competencies and gaps between academic preparation and workplace expectations.
- Conducted 32 semi-structured interviews with computing professionals to examine problem-solving and implicit Design Thinking practices.
- Analyzed qualitative and quantitative data and contributed to scholarly dissemination.

Researcher — Best Practices in HyFlex Learning (BPHE) Project — Purdue University | 2020–2021

- Contributed to research on HyFlex learning environments and students' academic and psychological needs.
- Supported analysis of student learning and experience data.

Researcher — Grievance Cell and Complaints Tracking System for Citizens | 2017–2018

- Developed and tested a mobile application for citizen grievance tracking and contributed to iterative design and data analysis.

Teaching Experience

Instructor — Think Like an Engineer, GERI Summer Program — Purdue University | Summers 2023, 2025 & 2026

- Designed and taught a two-week Design Thinking and engineering course for high school students.
- Created project-based learning experiences emphasizing human-centered problem solving, creativity, collaboration, prototyping, feedback, and iteration.

Teaching Assistant / Instructor — TECH 120: Design Thinking in Technology — Purdue University | 2020–2022

- Supported project-based undergraduate instruction in human-centered problem solving and iterative design.
- Facilitated team-based design activities, guided student projects, and provided feedback on learning and design work.

Workshop Co-Designer & Facilitator — Design Thinking / Website Design Workshop for High School Students — Purdue University | 2024

- Designed and facilitated hands-on learning experiences in persona development, low-fidelity prototyping, and user-centered design.

Course Instructor — Natural Disasters (Grades 3–4) | 2020

- Designed and taught inquiry-based science learning activities using experiments, models, simulations, and collaborative tasks.

Learning Design & Professional Experience

Instructional Designer Intern — FIRST Robotics Off-Season Programs — Purdue University | 2025–2026

- Conducted needs analysis through observations and interviews to inform an off-season robotics curriculum.
- Designed hands-on activities and instructional materials supporting problem definition, ideation, prototyping, iteration, and collaboration.
- Collaborated with program leaders and mentors to align curriculum with learner needs and program goals.

Graduate Assistant — IT Support, Graduate Office — Purdue University | 2024–Present

- Provide technical support for faculty, staff, and graduate students and support hardware/software systems.
- Develop instructional resources and provide training to support staff adoption of new technologies.

Peer Advisor — Career Services — Pittsburg State University | 2019–2020

- Supported students with CV/resume development, career resources, presentations, and academic/professional support.

Peer-Reviewed Publications

Tavakoli, J., Exter, M., & Kotangale, A. (2026). Integrating dispositions in instructional design: An exploratory study in higher education. Instructional Science, 54(1), Article 2. https://doi.org/10.1007/s11251-025-09762-3

Mentzer, N., Krishna, B., Kotangale, A., & Mohandas, L. (2023). HYFLEX environment: Addressing students' basic psychological needs. Learning Environments Research, 26(1), 271–289. https://doi.org/10.1007/s10984-022-09431-z

Kotangale, A., More, S., Sakharkar, P., Deshkar, D., & Banabakode, D. (2018). Grievance cell and complaints tracking system for citizens.

Manuscripts Under Review

Kotangale, A., Tavakoli, J., & Janakiraman, S. (under review). Investigating teaching methods in environmentally sustainable education within university engineering courses: A review of existing literature.

Kotangale, A., Exter, M., & Tagare, D. (under review). Computing professionals solving problems through a Design Thinking lens.

Kotangale, A., Tavakoli, J., & Exter, M. (under Review). Perspectives of computing and engineering educators on Design Thinking.

Kotangale, A., Exter, M., & Tavakoli, J. (under Review). Perspectives of computing and engineering educators on design models.

Manuscripts in Preparation

Kotangale, A., Yakut, N., & Exter, M. (in preparation). Understanding the Design Thinking mindset among adolescents in Grades 7–12.

Kotangale, A., Yakut, N., & Exter, M. (in preparation). Developing the Design Thinking mindset in high school students.

Technical Reports

Exter, M., Sabin, M., Janakiraman, S., Tagare, D., Kotangale, A., & Duan, S. (2023). DEAP competencies: Dispositions. DEAPening Employer Academic Partnerships.

Conference Presentations

Kotangale, A., Yakut, N., & Exter, M. (2026, November 3–7). Developing the Design Thinking mindset in high school students. AECT International Convention, Chicago, IL, United States. [Upcoming]

Kotangale, A., Yakut, N., & Exter, M. (2026, November 3–7).Kotangale, A., Yakut, N., & Exter, M. (in preparation). Exploring the development of Design Thinking mindsets in adolescent learners (Grades 7–12). AECT International Convention, Chicago, IL, United States. [Upcoming]

Tavakoli, J., Exter, M., & Kotangale, A. (2025, October). Integrating dispositions in instructional design: An exploratory study in higher education. AECT International Convention, Las Vegas, NV, United States.

Kotangale, A., Tavakoli, J., & Exter, M. (2024, October). Perspectives of computing and engineering educators on Design Thinking. AECT International Convention, Kansas City, MO, United States.

Tavakoli, J., Kotangale, A., Exter, M., Tagare, D., & Sabin, M. (2023, October). Dispositions in computing and implications for instructional design education. AECT International Convention, Orlando, FL, United States.

Kotangale, A., Tavakoli, J., & Janakiraman, S. (2023, October). Exploring pedagogical practices in environmentally sustainable education in university engineering courses: A literature review. AECT International Convention, Orlando, FL, United States.

Tavakoli, J., & Kotangale, A. (2023, April). Dispositions in instructional designing. AGSERS, Purdue University, West Lafayette, IN, United States.

Service & Leadership

Officer — Purdue Association of Learning Design & Technology (PALDT) — Purdue University | 2024–2025

- Collaborated with student leaders to support inclusive programming, student engagement, and professional-development activities.

Research & Technical Skills

Research Methods & Analysis: Quantitative and qualitative research; mixed-methods research; pre/post designs; descriptive and inferential statistics; qualitative coding; Dedoose; instrument adaptation; scenario-based assessment; rubric development; reliability analysis.

Learning Design: Curriculum development; needs analysis; project-based and experiential learning; workshop design; assessment design; learner-centered design; iterative development.

Technology & Facilitation: Educational technology integration; web/mobile development; workshop facilitation; student mentoring; collaborative learning; technology training.